

Statistical Hypothesis Testing Report

Factors Influencing Electric Vehicle Adoption

Team Name: ELECTRIC VS PETROL

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Course: Data Analysis Course

This document presents the statistical evaluation of the hypotheses used in the Electric Vehicle Adoption project. The analysis was conducted using Pearson Correlation, Linear Regression, t-tests, and F-tests with a significance level of $\alpha = 0.05$.

H1: Fuel Prices Increase EV Adoption

H0: Fuel prices do not significantly affect EV market share.

H1: Higher fuel prices significantly increase EV market share.

Correlation (r): 0.554

t-statistic: 23.021

p-value: 2.15×10^{-97}

R²: 0.307

F-statistic: 529.952

Decision: Reject H0

Result: Supported

Interpretation: Fuel prices have a statistically significant positive impact on EV adoption.

H2: Charging Stations Increase EV Adoption

H0: Charging stations do not significantly affect EV market share.
H1: More charging stations significantly increase EV market share.

Correlation (r): 0.214
t-statistic: 7.573
p-value: 7.24×10^{-14}
R²: 0.046
F-statistic: 57.357
Decision: Reject H0
Result: Supported

Interpretation: Charging infrastructure significantly improves EV market share.

H3: Government Subsidies Increase EV Adoption

H0: Government subsidies do not significantly affect EV market share.
H1: Government subsidies positively influence EV adoption.

Correlation (r): -0.056
t-statistic: -1.929
p-value: 0.054
R²: 0.003
F-statistic: 3.721
Decision: Fail to Reject H0
Result: Not Supported

Interpretation: Government subsidies showed weak statistical significance in this dataset.

H4: GDP per Capita Increases EV Adoption

H0: GDP per capita does not significantly affect EV market share.
H1: Higher GDP per capita significantly increases EV adoption.

Correlation (r): 0.412
t-statistic: 15.640
p-value: 2.60×10^{-50}
R²: 0.170

F-statistic: 244.595

Decision: Reject H0

Result: Supported

Interpretation: GDP per capita has a statistically significant positive relationship with EV market share.

Final Conclusion

The statistical analysis confirms that fuel prices, charging infrastructure, and GDP per capita significantly influence EV adoption. Government subsidies did not show statistically significant evidence at the 5% significance level. The combined regression analysis demonstrates that EV adoption is best explained through multiple interacting variables.